

The raw data exported from the survey software included metadata rows and 22 incomplete responses. I utilized Pandas to clean the dataset, standardizing column names and isolating responses required for accurate statistical analysis. The target of $N = 25$ complete responses is being investigated as the initial cleaning steps were too aggressive.

Import all necessary libraries.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats
import numpy as np
```

The data I used for this project came from a Qualtrics survey. Qualtrics has two rows of metadata that need to be handled first. I used Python to create a "cheat sheet" of current column names and their associated survey questions, then used it to create new column names. The naming convention varied by question type: demographic questions were named based on the demographic being measured, and I used `inv_` or `sat_` with the associated activity to create involvement and satisfaction names. `inv` = involvement, `sat` = satisfaction, `fre` = frequency, `dur` = duration

```
#Load the raw data from a .csv file.
df_raw = pd.read_csv('oap_data.csv')
# Data comes from a qualtrics survey, so the first two rows are survey metadata.

#df_clean = df.iloc[2:].copy()
print("--- COLUMN DICTIONARY CHEAT SHEET ---")
for col in df_raw.columns:
    # df_raw[col].iloc[0] grabs the first row of data (the Qualtrics metadata/question text)
    print(f"Column: {col} --> Question: {df_raw[col].iloc[0]}")
```

Column: Q98_1 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Hiking/Walking

Column: Q98_2 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Trail Running

Column: Q98_3 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Picnicking

Column: Q98_12 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Biking

Column: Q98_13 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Fishing

Column: Q98_14 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Swimming

Column: Q98_15 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Kayaking

Column: Q98_16 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Canoeing

Column: Q98_17 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Paddle Boarding

Column: Q98_18 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Camping

Column: Q98_19 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Zipline

Column: Q98_20 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Challenge Course

Column: Q98_21 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Laying on the Beach

Column: Q98_22 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Disc Golf

Column: Q98_23 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Nature Viewing

Column: Q98_24 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Photography

Column: Q98_25 --> Question: 6. Please indicate your frequency of daily involvement over a month in the following LU Outdoor Recreation activities, during a typical semester (For example, rating 20 for Trail Running would mean you run 20 days per month during a typical semester.) - Other

Column: Q97 --> Question: Please specify other activity.

Column: Q99 --> Question: How long have you been visiting the LU Outdoor Recreation setting/facilities?

Code to change column names using a dictionary:

```
rename_dict = {
    'Q1a' : 'inv_mobile_media',
    'Q1b' : 'inv_mobile_media_freq',
    'Q1c_1' : 'inv_mobile_media_dur',
    'Q1d' : 'inv_mobile_media_people',
    'Q1e' : 'sat_mobile_media',
    'Q2a' : 'inv_home',
    'Q2b' : 'inv_home_freq',
    'Q2c_1' : 'inv_home_dur',
    'Q2d' : 'inv_home_people',
    'Q2e' : 'sat_home',
    'Q3a' : 'inv_chores',
    'Q3b' : 'inv_chores_freq',
    'Q3c_1' : 'inv_chores_dur',
    'Q3d' : 'inv_chores_people',
    'Q3e' : 'sat_chores',
    'Q4a' : 'inv_nature',
    'Q4b' : 'inv_nature_freq',
    'Q4c_1' : 'inv_nature_dur',
    'Q4d' : 'inv_nature_people',
}
```

'Q4e' : 'sat_nature',
'Q5a' : 'inv_home_physical',
'Q5b' : 'inv_home_physical_freq',
'Q5c_1' : 'inv_home_physical_dur',
'Q5d' : 'inv_home_physical_people',
'Q5e' : 'sat_home_physical',
'Q6a' : 'inv_org',
'Q6b' : 'inv_org_freq',
'Q6c_1' : 'inv_org_dur',
'Q6d' : 'inv_org_people',
'Q6e' : 'sat_org',
'Q7a' : 'inv_park',
'Q7b' : 'inv_park_freq',
'Q7c_1' : 'inv_park_dur',
'Q7d' : 'inv_park_people',
'Q7e' : 'sat_park',
'Q8a' : 'inv_club',
'Q8b' : 'inv_club_freq',
'Q8c_1' : 'inv_club_dur',
'Q8d' : 'inv_club_people',
'Q8e' : 'sat_club',
'Q9a' : 'inv_spec',
'Q9b' : 'inv_spec_freq',
'Q9c_1' : 'inv_spec_dur',
'Q9d' : 'inv_spec_people',
'Q9e' : 'sat_spec',
'Q10a' : 'inv_comm',
'Q10b' : 'inv_comm_freq',
'Q10c_1' : 'inv_comm_dur',
'Q10d' : 'inv_comm_people',
'Q10e' : 'sat_comm',
'Q11a' : 'inv_sum_camp',
'Q11b' : 'inv_sum_camp_freq',
'Q11c_1' : 'inv_sum_camp_dur',
'Q11d' : 'inv_sum_camp_people',
'Q11e' : 'sat_sum_camp',
'Q12a' : 'inv_camping',
'Q12b' : 'inv_camping_freq',
'Q12c_1' : 'inv_camping_dur',
'Q12d' : 'inv_camping_people',
'Q12e' : 'sat_camping',
'Q13a' : 'inv_hunfish',
'Q13b' : 'inv_hunfish_freq',
'Q13c_1' : 'inv_hunfish_dur',
'Q13d' : 'inv_hunfish_people',
'Q13e' : 'sat_hunfish',
'Q14a' : 'inv_water',
'Q14b' : 'inv_water_freq',
'Q14c_1' : 'inv_water_dur',
'Q14d' : 'inv_water_people',
'Q14e' : 'sat_water',
'Q15a' : 'inv_water_motor',
'Q15b' : 'inv_water_motor_freq',
'Q15c_1' : 'inv_water_motor_dur',
'Q15d' : 'inv_water_motor_people',
'Q15e' : 'sat_water_motor',
'Q16a' : 'inv_adventure',
'Q16b' : 'inv_adventure_freq',

```

'Q16c_1' : 'inv_adventure_dur',
'Q16d' : 'inv_adventure_people',
'Q16e' : 'sat_adventure',
'Q17a' : 'inv_atv',
'Q17b' : 'inv_atv_freq',
'Q17c_1' : 'inv_atv_dur',
'Q17d' : 'inv_atv_people',
'Q17e' : 'sat_atv',
'Q18a' : 'inv_tourism',
'Q18b' : 'inv_tourism_freq',
'Q18c_1' : 'inv_tourism_dur',
'Q18d' : 'inv_tourism_people',
'Q18e' : 'sat_tourism',
'Q91' : 'gender',
'Q92' : 'age',
'Q93' : 'class_rank',
'Q94' : 'major',
'Q95' : 'athletic_participation',
'Q98_1' : 'activity_hikewalk',
'Q98_2' : 'activity_trailrun',
'Q98_3' : 'activity_picnick',
'Q98_12' : 'activity_bike',
'Q98_13' : 'activity_fish',
'Q98_14' : 'activity_swim',
'Q98_15' : 'activity_kayak',
'Q98_16' : 'activity_canoes',
'Q98_17' : 'activity_paddboard',
'Q98_18' : 'activity_camp',
'Q98_19' : 'activity_zipline',
'Q98_20' : 'activity_chaircourse',
'Q98_21' : 'activity_beach',
'Q98_22' : 'activity_discgolf',
'Q98_23' : 'activity_natureview',
'Q98_24' : 'activity_photo',
'Q98_25' : 'activity_other',
'Q99' : 'lu_facility_freq'
}

```

#Apply the new column names to the database.

```
df = df_raw.copy() # Start with a copy of df_raw
```

```
original_cols_list = df.columns.tolist()
```

```
print(f"Original df_raw columns (first 5): {original_cols_list[:5]}")
```

```
df.columns = df.columns.str.strip() # Strip whitespace from column names for robust renaming
```

```
stripped_cols_list = df.columns.tolist()
```

```
print(f"Stripped columns (first 5): {stripped_cols_list[:5]}")
```

Check which columns from rename_dict were not found in the actual DataFrame

```
missing_keys = [key for key in rename_dict.keys() if key not in stripped_cols_list]
```

```
if missing_keys:
```

```
    print(f"Warning: The following rename_dict keys were NOT found in the DataFrame's columns: {missing_keys[:5]}...")
```

Check which 'Q' columns were in the DataFrame but NOT in rename_dict, and therefore not renamed

```
unrenamed_q_cols = [col for col in stripped_cols_list if col.startswith('Q') and col not in rename_dict.keys()]
```

```
if unrenamed_q_cols:
```

```
    print(f"Warning: The following 'Q' columns were in the DataFrame but NOT in rename_dict (first 5): {unrenamed_q_cols[:5]}...")
```

```
df = df.rename(columns=rename_dict)

renamed_cols_list = df.columns.tolist()
print(f"Columns AFTER renaming (first 5 and last 5): {renamed_cols_list[:5]}...{renamed_cols_list[-5:]}")

df_clean = df.drop([0, 1]).reset_index(drop=True)
display(df_clean.head())
```

Original df_raw columns (first 5): ['Q1a', 'Q1b', 'Q1c_1', 'Q1d', 'Q1e']

Stripped columns (first 5): ['Q1a', 'Q1b', 'Q1c_1', 'Q1d', 'Q1e']

Warning: The following 'Q' columns were in the DataFrame but NOT in rename_dict (first 5): ['Q97']...

Columns AFTER renaming (first 5 and last 5): ['inv_mobile_media', 'inv_mobile_media_freq', 'inv_mobile_media_dur', 'inv_mobile_media_people', 'sat_mobile_media']...['activity_

	inv_mobile_media	inv_mobile_media_freq	inv_mobile_media_dur	inv_mobile_media_people	sat_mobile_media	inv_home	inv_home_freq	inv_home_dur	inv_home_people	sat_home
0	2	NaN	NaN	NaN	NaN	2	NaN	NaN	NaN	NaN
1	1	1	4	3	3	1	3	3	3	3
2	1	1	2	3	4	1	1	2	3	5
3	1	1	1	1	4	1	3	2	3	4
4	1	1	1	3	4	1	2	1	3	4

5 rows × 114 columns

After changing the columns names I proceeded with further cleaning the dataset by removing rows if atleast one column has a null value.

```
# Identify involvement and satisfaction columns
inv_cols = [col for col in df_clean.columns if col.startswith('inv_')]
sat_cols = [col for col in df_clean.columns if col.startswith('sat_')]

# Drop rows only if they left a demographic questions blank
demographics = ['gender', 'age', 'class_rank']
df_clean = df_clean.dropna(subset=demographics)

# Change involvement and satisfaction columns from text to numbers
# Use errors='coerce' to turn any weird text artifacts into NaNs safely
df_clean[inv_cols] = df_clean[inv_cols].apply(pd.to_numeric, errors='coerce')
df_clean[sat_cols] = df_clean[sat_cols].apply(pd.to_numeric, errors='coerce')

# Drop any rows that became completely blank in our target columns
df_clean = df_clean.dropna(subset=inv_cols + sat_cols, how='all')

# Math to find total involvement score and total satisfaction score
df_clean['Total_Involvement'] = df_clean[inv_cols].sum(axis=1)
df_clean['Total_Satisfaction'] = df_clean[sat_cols].sum(axis=1)
```

Graph the results on a scatter plot.

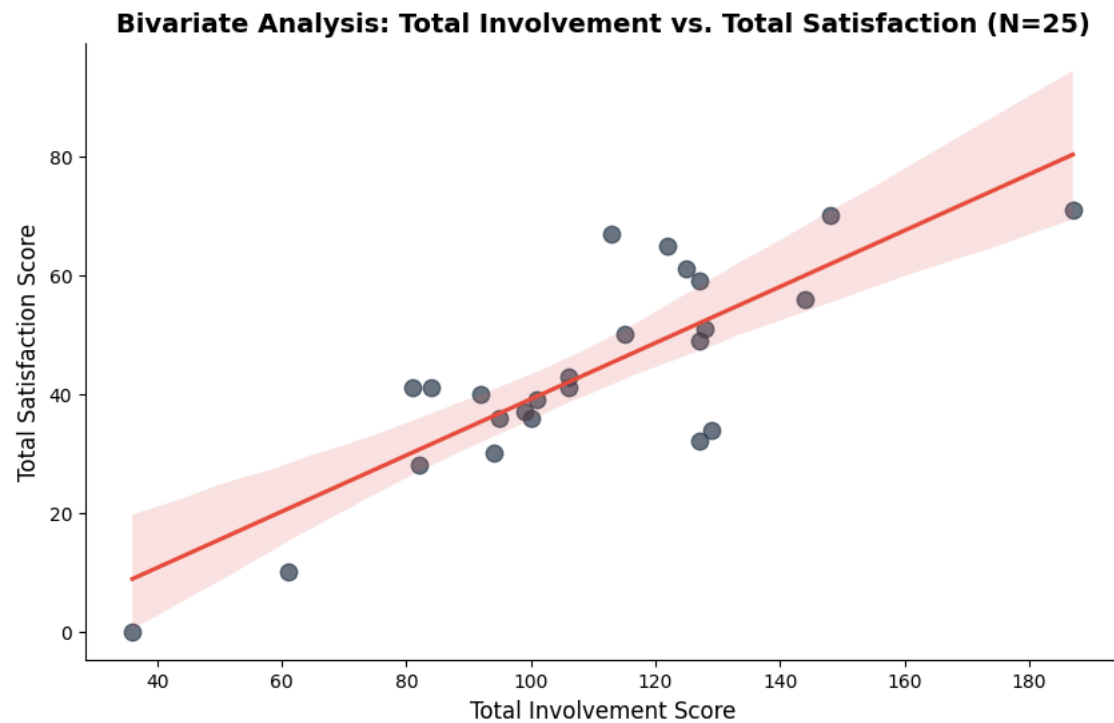
```
plt.figure(figsize=(10, 6))

sns.regplot(
    data=df_clean,
    x='Total_Involvement',
    y='Total_Satisfaction',
    scatter_kws={'alpha': 0.7, 's': 80, 'color': '#2c3e50'},
    line_kws={'color': '#e74c3c'}
)

plt.title('Bivariate Analysis: Total Involvement vs. Total Satisfaction (N=25)', fontsize=14, fontweight='bold')
plt.xlabel('Total Involvement Score', fontsize=12)
plt.ylabel('Total Satisfaction Score', fontsize=12)

sns.despine()

plt.show()
```



Math to find total involvement score and total satisfaction score. Scatter plot graph to show the bivariate analysis.

Code to run a T test on male versus female satisfaction.

```
# Split the data into the two groups
males = df_clean[df_clean['gender'] == '1']['Total_Satisfaction']
females = df_clean[df_clean['gender'] == '2']['Total_Satisfaction']
```

```
# Independent t-test
t_stat, p_value = stats.ttest_ind(males, females)

print(f"T-statistic: {t_stat:.3f}")
print(f"P-value: {p_value:.3f}")

# Create a visual Box Plot
sns.boxplot(data=df_clean, x='gender', y='Total_Satisfaction')
plt.title('Satisfaction Scores by gender')
```

T-statistic: -0.740
P-value: 0.467

